

PARV RAJ CHAUHAN

Pre-Final Year CS Student · AI Engineering

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Summary

Pre-final year CS student at IIIT Nagpur with hands-on experience shipping production-grade backend systems and ML inference pipelines. Built 3+ end-to-end projects across FastAPI microservices, Transformer fine-tuning (DeBERTa-v3), dual-index RAG pipelines, and Dockerised cloud deployments with full SDLC ownership from schema design to CI/CD. Strong foundation in DSA and systems; focused on the intersection of scalable APIs and applied NLP/AI.

Skills

Languages	Python, C++, SQL, JavaScript
Backend & APIs	FastAPI, Node.js, Express, REST APIs, Microservices Architecture
Databases	PostgreSQL, MongoDB, MySQL, FAISS (Vector Database), Relational Schema Design
ML & AI	PyTorch, Hugging Face Transformers, Scikit-learn, MLflow, RAG, Embeddings, LLM Integration, LangChain, LangGraph (<i>actively learning</i>), SHAP Explainability
DevOps & Infra	Docker, Docker Compose, GitHub Actions (CI/CD), Linux, AWS (S3/EFS), Render
Frontend	React, Next.js, Tailwind CSS
CS Fundamentals	Data Structures & Algorithms (DSA), OOP, DBMS, Operating Systems, Computer Networks

Projects

GUID: AI-Powered Resume Analysis Platform [Link](#) *Python, Node.js, Express, MongoDB, React, Docker, Render*

- Designed and built scalable REST APIs in Node.js/Express following microservices principles, handling PDF parsing, job description analysis, and AI report generation via Gemini API.
- Implemented full backend architecture authentication, MongoDB data modeling, and structured API responses.
- Orchestrated a Docker Compose setup with environment isolation; deployed frontend and backend as services on Render.
- Maintained end-to-end SDLC ownership: feature scoping, modular code, endpoint testing (Postman), mobile-responsive React frontend and incremental shipping closely mirroring professional engineering workflows.

RAG Movie Q&A System [Link](#) *Python, FastAPI, FAISS, sentence-transformers, Groq, Next.js*

- Built a production-oriented RAG pipeline with section-aware chunking (title/cast/director/plot typed chunks) and movie-level score aggregation, eliminating cross-film context mixing during generation.
- Implemented constrained LLM generation grounded strictly in retrieved context with Precision@K and Recall@K evaluation metrics; documented known failure modes (ambiguous titles, franchise queries) for iterative improvement.
- Designed an end-to-end inference pipeline : query embedding → FAISS top-K retrieval → score aggregation → context selection serving a Groq-backed LLM constrained to retrieved context only, with FastAPI exposing the pipeline.

AI-Based Web Application Firewall : Anomaly Detection [Link](#) *Python, PyTorch, DeBERTa-v3, FAISS, FastAPI, Docker*

- Built pipeline: NGINX log generation, parsing, normalization, and Hugging Face tokenization, processing 130K+ HTTP requests.
- Fine-tuned DeBERTa-v3-base with self-supervised MLM on HTTP traffic; implemented dual PLL scoring: Exact (100% precision, 0% FPR) and Fast (80% F1, 11.54% FPR); high-confidence scores(9.5) trigger SHAP explainability.
- Designed and implemented a hybrid RAG retrieval system with dual FAISS indexes (HNSW for 44K CVE vectors, FlatIP for 1.9K CWE/OWASP vectors) achieving 100% keyword recall on matched queries and a 10/10 hallucination-free generation score.
- Exposed the WAF and RAG pipeline via FastAPI endpoints and containerized the full service using Docker; implemented CI/CD automation to rebuild FAISS indexes on new CVE/CWE ingestion, maintaining an LLM-as-judge weighted score of 8.7/10 across groundedness, citation correctness, and relevance dimensions.

Education

Indian Institute of Information Technology (IIIT), Nagpur
B.Tech in Computer Science and Engineering CGPA: **8.69 / 10.0**

Expected June 2027
Maharashtra, India

Delhi Public School, Aligarh
Class XII: 91.0% & Class X: 95.8%

2022

Achievements

- Competitive Programming:** CodeChef **3-Star** (PR: **1608**)
- Hackathons:** **2nd Place** at **StratQuest** — **SHARE** × **IIT Kharagpur Spring Fest** (ML–Business track); **two-time** qualifier (**2024, 2025**) in **Smart India Hackathon (SIH)** internal rounds.